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Near-surface distributions of soil water and water repellency under three effluent irrigation schemes in a blue gum (*Eucalyptus globulus*) plantation

L.A. Thwaites^a, G.H. de Rooij^{b,*}, S. Salzman^c, G. Allinson^{a,1}, F. Stagnitti^a, R. Carr^c, V. Versace^a, S. Struck^d, T. March^a

^a School of Ecology and the Environment, Deakin University, P.O. Box 423, Warrnambool, Victoria 3280, Australia

^b Department of Environmental Sciences, Soil Physics, Ecohydrology, and Groundwater Management Group, Wageningen University, Nieuwe Kanaal 11, 67090 PA Wageningen, The Netherlands

^c School of Information Systems, Deakin University, P.O. Box 423, Warrnambool, Victoria 3280, Australia

^d Institute for Land Use, University of Rostock, Justus-von-Liebig-Weg 6, D-18059 Rostock, Germany

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ABSTRACT

Water repellent soils are difficult to irrigate and susceptible to preferential flow, which enhances the potential for accelerated leaching to groundwater of hazardous substances. Over 5 Mha of Australian soil is water repellent, while treated municipal sewage is increasingly used for irrigation. Only if a critical water content is exceeded will repellent soils become wettable. To avoid excessive loss of water from the root zone via preferential flow paths, irrigation schemes should therefore aim to keep the soil wet enough to maintain soil wettability. Our objective was to monitor the near-surface water content and water repellency in a blue gum (*Eucalyptus globulus*) plantation irrigated with treated sewage. The plantation's sandy soil surface was strongly water repellent when dry. For 4 months, three rows of 15 blue gum trees each received no irrigation, three other rows received 50% of the estimated potential water use minus rainfall, and three more rows received 100%. During this period, 162 soil samples were obtained in three sampling rounds, and their water content (% dry mass) and degree of water repellency determined. Both high and low irrigation effectively wetted up the soil and eliminated water repellency after 2 (high) or 4 (low) months. A single-peaked distribution of water contents was observed in the soil samples, but the water repellency distribution was dichotomous, with 44% extremely water-repellent and 36% wettable. This is consistent with a threshold water content at which a soil sample changes from water repellent to wettable, with spatial variability of this threshold creating a much wider transition zone at the field scale. We characterized this transition zone by expressing the fraction of wettable samples as a function of water content, and demonstrated a way to estimate from this the wettable portion of a field from a number of water content measurements. To keep the plantation soil wettable, the water content must be maintained at a level at which a significant downward flux is likely, with the associated enhanced leaching. At water contents with negligible downward flux, the field is water repellent, and leaching through preferential flow paths is likely. Careful management is needed to resolve these conflicting requirements.

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* Corresponding author. Tel.: +31 317 482778/485311; fax: +31 317 484885.

E-mail address: Ger.deRooij@wur.nl (G.H. de Rooij).

¹ Present address: Primary Industries Research Victoria, Department of Primary Industries Queenscliff Centre, P.O. Box 114, Queenscliff, Victoria 3225, Australia.

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1. Introduction

For well over a century, soil water repellency, or the resistance of soil to instantaneous wetting, has been a concern of both scientists and land managers (DeBano, 1969, 2000), and is now an acknowledged multi-disciplinary aspect of soil science (DeBano, 2000). Over the last decade water repellency has been reported for varying soils, land uses, and climatic conditions on all inhabited continents (e.g. Dekker et al., 1998; Blackwell, 2000; DeBano, 2000; Dekker and Ritsema, 2000; Doerr et al., 2000; Keizer et al., 2005a; Moral Garcia et al., 2005). With a rapidly growing body of literature, it is becoming increasingly evident that soil water repellency is a widespread phenomenon (DeBano, 1969, 2000; Doerr et al., 2000; Keizer et al., 2005b; Ritsema and Dekker, 2005). For instance, in Australia, numerous predominantly sandy regions with a combined area of at least 5 Mha are water repellent (Bond and Harris, 1964; Roberts, 1966; Harper and Gilkes, 1994; Blackwell, 2000; McKissock et al., 2000).

Soil water repellency is generally believed to be caused by long-chained hydrophobic organic compounds present as interstitial matter or coating soil mineral or aggregate surfaces (Doerr et al., 2000; Franco et al., 2000; Dekker et al., 2001; Morley et al., 2005; Xiong et al., 2005b). These hydrophobic compounds are derived from living or decomposing plants, plant root exudates, decomposing soil organic matter, fungal residues, resins, waxes, aromatic oils, and soil microorganisms (McGhie and Posner, 1981; Mallik and Rahman, 1985; Bisdorf et al., 1993; Dekker and Ritsema, 1994; Doerr et al., 1998; DeBano, 2000; Doerr et al., 2000; McKissock et al., 2000).

Variations in soil water content can greatly affect the water repellent status of a soil (see overview by Doerr et al., 2000). Many authors reported an inverse relationship between soil water content and soil water repellency (e.g. Dekker and Ritsema, 1994; Dekker et al., 1998; Doerr et al., 2000; Coelho et al., 2005; Keizer et al., 2005a; Leighton-Boyce et al., 2005). Dekker and Ritsema (1994) proposed that soils will remain water repellent until a critical soil water content is reached above which the soils become wettable. Recently, Dekker et al. (2001) hypothesized that the repellency change might occur over a broader soil water range, termed the 'transition zone'.

Eucalyptus spp. plantations produce some of the most severe repellency patterns observed in the world (Doerr et al., 2000; Coelho et al., 2005; Keizer et al., 2005a; Leighton-Boyce et al., 2005). Possibly, plants might use hydrophobic substances to suppress the germination of competing vegetation, and to increase water conservation by channelling water deep into the soil profile along preferential flow pathways, thereby reducing evaporation (Scott, 1992; Moore and Blackwell, 1998; Doerr et al., 2000).

Compared to many countries, Australia has rather limited and relatively poor quality freshwater resources. Recent regulations discouraged the use of groundwater and river water for irrigation, forcing many farmers to seek alternative water sources, such as recycled treated municipal sewage and storm water (Myers et al., 1999; Xiong et al., 2005a). A recent study indicated that prolonged (>20 years) irrigation of wettable soils with treated sewage caused considerable water repellency around the irrigation dripper outlets (Wallach et al.,

2005). The effects on soils that are already water repellent before irrigation with sewage are unknown.

The potential for accelerated leaching to groundwater of hazardous chemicals and pathogens via preferential flow is a serious concern for artificial and natural groundwater recharge (see Bouwer, 2002) and conservation of groundwater as a source of potable water (Bouwer, 1978, pp. 407–418). This is a common problem in water repellent soils and requires careful management (Blackwell, 2000), especially since soil salinization can only be prevented by a time-averaged downward water flux through and beyond the root zone (van Schilfgaarde, 1984). Ritsema and Dekker (2000) state that in order to manage water repellent soil systems in a sustainable manner the incidence of irregular wetting and preferential flow should be prevented or combated. Careful scheduling of irrigation timing and quantities may help prevent soils from becoming water repellent by keeping the soil water contents above critical levels (Dekker et al., 2001; Ritsema et al., 2005). Success of this approach depends largely on the constant monitoring of soil water content at different depths in the profile, and on the timely application of sufficient amounts of irrigation water. Such an approach would guarantee uniform infiltration and prevent preferential flow and transport from occurring thus reducing the leaching risks to subsoil environments (Ritsema et al., 2005).

Irrigation of tree plantations, in particular *Eucalyptus* spp., is becoming a popular alternative to river or ocean discharge of treated sewage effluent (Myers et al., 1996; Falkiner and Smith, 1997; Myers et al., 1999). There is considerable experience with the use of treated sewage for irrigation (e.g. Feigin et al., 1991), but little is known about the risks and benefits of water repellent soils often found in tree plantations. Irrigating with recycled water may increase plantation growth through increased nutrient loading provided the associated environmental risks of contamination of soils and groundwater can be minimized (Xiong et al., 2005a), and the development of severe repellency patterns minimized. The specific aim of this study, therefore, was to monitor the in-field response of soil water content and soil water repellency to wetting by municipal effluent irrigation under an established blue gum (*Eucalyptus globulus*) plantation, and to assess the implications for irrigation management.

2. Materials and methods

2.1. Experimental site

The experimental site was located at the Deans Heath Agroforestry trial site owned by Portland Aluminium 6 km south east of the city of Portland, Victoria, Australia (38°39'S, 141°61'E) (Fig. 1). Situated within the 600 ha of heath lands and wetlands, the experimental plot was 48 ± 1.16 m above sea level, and sloping downward by approximately 6% from the South East to the North West. Long-term mean annual rainfall is 795.5 mm, with a winter dominant distribution. Mean minimum temperature of the coolest month is 6.1 °C (July) and the mean maximum temperature for the hottest month is 22.7 °C (February). Annual class A pan evaporation is 1176.5 mm, with monthly net evaporation exceeding 50 mm

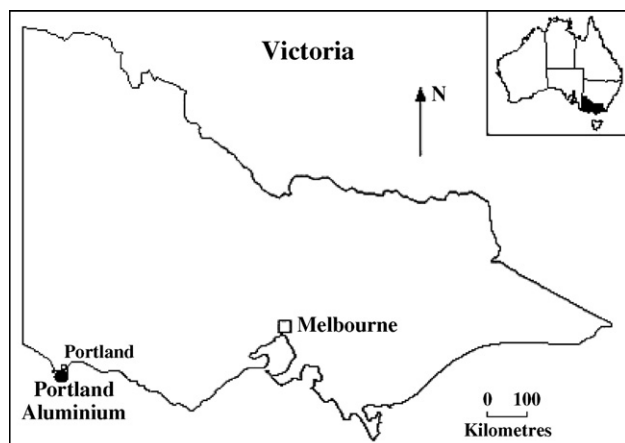


Fig. 1 – Portland Aluminium site 6 km south east of the city of Portland, Western Victoria, Australia (38°39'S, 141°61'E).

from September through May. In, 1998, the site was planted with *E. globulus* (tree spacing 2 m, row spacing 4 m).

2.2. Soils

The site has a sandy soil (USDA classification; [Gee and Or, 2002](#)) with less than 2% silt and clay at any of the sampled depth intervals (0–1 cm; 1–16.3 cm; 16.3–50 cm). The surface layer (0–16.3 cm interval) is strongly water repellent with 96.3% of water drop penetration times (WDPT) exceeding 600 s for dry soils. Repellency is variable in the 16.3–30.7 cm interval with approximately 50% of samples repellent and 50% of samples non-repellent. The 33.7–50 cm interval is predominantly wettable with 77.8% of WDPTs recorded as <5 s. Further selected soil properties are given in [Table 1](#).

2.3. Experimental design

High, low, and control irrigation treatments were randomly allocated across nine rows of the experimental plot ([Fig. 2](#)). From historic rainfall records and pan evaporation data from a station a few kilometres away ([Laidlaw, 2000](#)) we estimated an average representative rainfall rate of 1.1 mm day⁻¹, and a pan evaporation rate of 5.2 mm day⁻¹. [Morris and Wehner \(1987\)](#) presented crop factors for young eucalypt stands with non-closed canopies (as ours) between 1.4 and 1.9. We combined our tree spacing (8 m² of soil surface per tree) with the lower value for the crop factor to arrive at an average daily evaporation deficit of 0.05 m³ day⁻¹ per tree. The high treatment (0.048 m³ day⁻¹ irrigation per tree) aimed to

compensate for the deficit. The low treatment provided 0.024 m³ day⁻¹ per tree. The controls received no irrigation. Trees were irrigated with secondary treated municipal effluent, which was supplied by Portland Coast Water (now Wannon Water) to Deans Heath Agroforestry trial site via an automated system. Each irrigated tree received water from either one (low treatment) or two (high treatment) drippers, each delivering 0.08 m³ h⁻¹, between midnight and 3.00 a.m. every 24 h during the entire irrigation period.

To randomise dripper placement a plastic hoop (1 m diameter) marked as a clock face was placed around each tree, with 12 o'clock facing South. A virtual transect was run from the base of the tree to a randomly selected time interval (e.g. 1 o'clock). For the high treatment, we placed two drippers delivering 8 L h⁻¹ along the virtual transect half way between the tree and the circumference of the hoop, one dripper (8 L h⁻¹) for the low treatment, and no drippers were used for the controls. Control sampling points were also defined in this manner. For this study six trees (Tree Nos. 1, 3, 6, 9, 12, 15) in each of the nine rows were systematically selected for dripper installation and/or sampling.

2.4. Soil sampling and in-field assessment

To assess the in-field response of soil water repellency to wetting by effluent irrigation, the repellency status and the soil water content of the site were measured *in situ* on three occasions over a 4-month period in the Australian spring, summer and autumn of 2003–2004 (27 November 2003, 11 February 2004, 25 March 2004). [Leighton-Boyce et al. \(2005\)](#) suggested that the existence of an apparent transition zone may be an artefact of the different spatial scales at which repellency and soil water content are assessed. Since soil water contents cannot be determined for the minute volumes of soil with which infiltrating droplets make contact during a wettability measurement, we minimized the discrepancy in the vertical observation scale by only collecting soil material from a thin layer at the soil surface for the water content measurements.

At each selected tree, we removed leaf litter and humus by hand from a 10 cm × 10 cm area. For the high and low treatments measurements were taken from immediately below the drippers. Samples for soil water analysis (% dry mass) were taken by scrapping a small garden spade across the surface layer (<1 cm depth) adjacent to the sampling point. Samples were stored in sealed plastic bags until laboratory analyses. Soil water was assessed according to [Briggs \(1977\)](#). On the second and third sampling trips, the samples were taken from a point directly adjacent to the initial sampling point.

Table 1 – Selected soil properties of Deans Heath Agro forestry trial site

Depth interval (m)	Mean bulk density (S.D.) ^a (kg m ⁻³)	Estimated saturated hydraulic conductivity ^b (m day ⁻¹)
0–0.163	1.07 × 10 ³ (1.7 × 10 ²)	
0.163–0.337	1.41 × 10 ³ (1.3 × 10 ²)	8.14
0.337–0.500	1.48 × 10 ³ (70)	8.90

^a According to [Briggs \(1977\)](#).

^b According to [Alyamani and Sen \(1993\)](#).

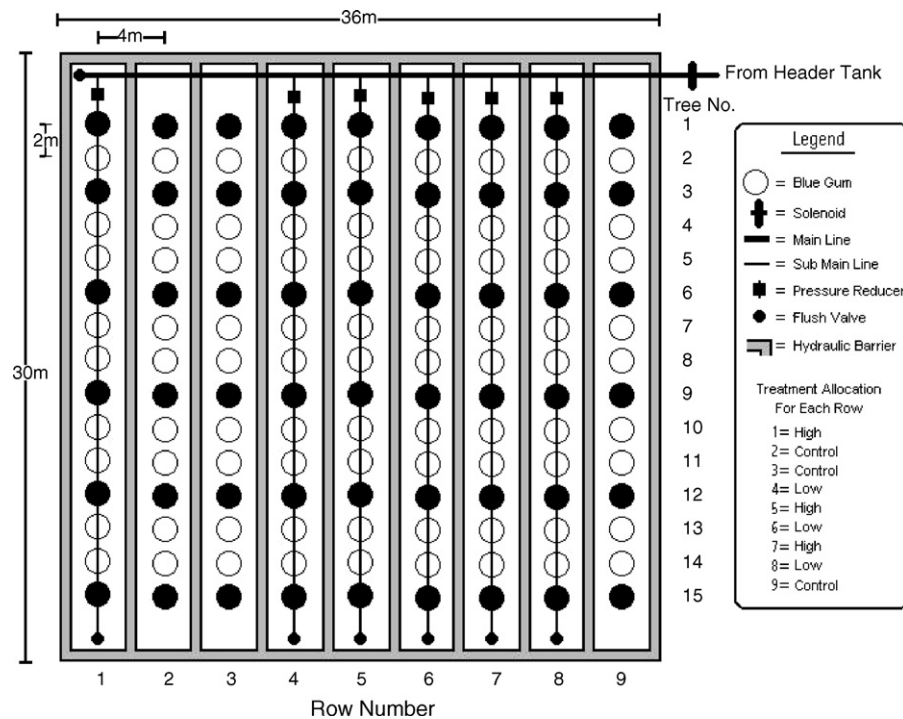


Fig. 2 – Experimental set-up and irrigation system design in an established *Eucalyptus globulus* plantation. Rows 1, 5, and 7 received the estimated potential evapotranspiration minus rainfall (high irrigation); rows 4, 6, and 8 received 50% of the high irrigation (low irrigation), and control rows 2, 3, and 9 received no irrigation. Trees below which samples were taken are blacked out.

Table 2 – Properties of ethanol–water mixtures used to assess soil water repellency (based on the molarity of ethanol droplet method (see Doerr, 1998; Leighton-Boyce et al., 2005), and associated descriptive water repellency severity classes as defined by the lowest ethanol concentration at which three of five droplets will infiltrate within 3 s)

Ethanol concentration (vol.%)	Surface tension ($\times 10^{-3} \text{ N m}^{-1}$)	Severity class	Descriptive category
0.0	72.1	1	Wettable
1.0	66.9	2	Low repellency
2.0	63.3	3	
3.0	60.9	4	
5.0	56.6	5	Moderate repellency
8.5	51.2	6	
13	46.3	7	
18	42.3	8	Severe repellency
24	38.6	9	
36	33.1	10	
50	31.0	11	Extreme repellency
>50	<31.0	12	

At each sampling point, soil water repellency was measured *in situ* at the soil surface. We assessed the actual water repellency (Dekker and Ritsema, 1994) of the field moist soil surface by the molarity of an ethanol droplet (MED) method (Doerr, 1998; Leighton-Boyce et al., 2005). The lowest concentration of ethanol at which at least three of five applied ethanol/water drops infiltrate within 3 s characterizes the severity of repellency in terms of 12 classes (class 1: 0% ethanol (wettable); class 12: >50% ethanol by volume (extreme repellency)) (Doerr, 1998; Leighton-Boyce et al., 2005). The ethanol concentrations, their surface tension and associated descriptive severity classes used in this study have been adapted from Doerr (1998) and Leighton-Boyce et al. (2005) and are given in Table 2. Once the severity of water repellency was determined at each sampling point, the litter layer was replaced.

3. Results and discussion

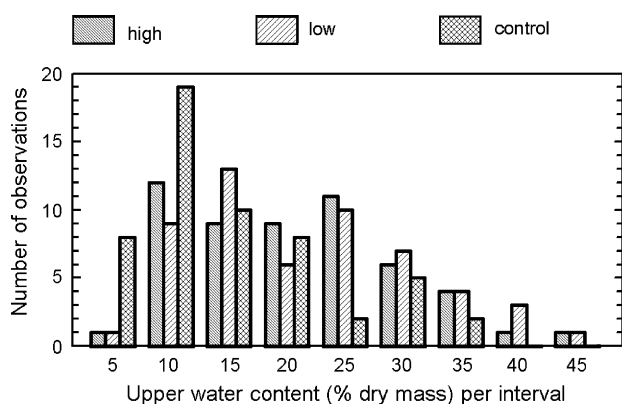
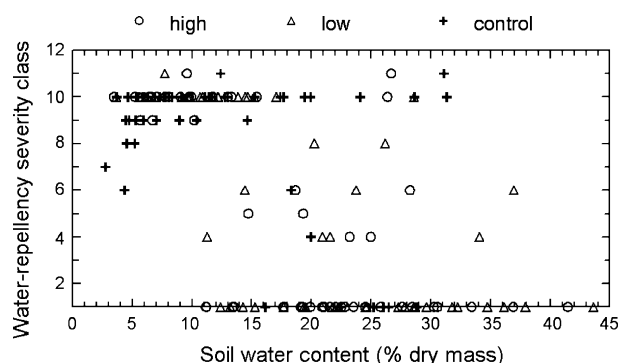
Table 3 shows the seasonal changes in soil water and development of soil water repellency in the irrigated and control treatments, as calculated from the median ethanol concentration from each treatment. Pre-irrigation sampling (27 November 2003), post winter but after the soils had had a short period to dry, shows that all treatments exhibited extreme repellency (severity class 10—Table 2) and had relatively low soil water contents, even though there had been 27 mm of rainfall 1 week earlier.

Table 3 – Median water repellency severity class (WR; see Table 2) measured in situ and near-surface soil water content (SW) for the various irrigation treatments at three dates

Treatment	27 November 2003		11 February 2004		25 March 2004	
	WR	SW (% dry mass)	WR	SW (% dry mass)	WR	SW (% dry mass)
Control	10	9.7	9	17.8	10	5.4
Low	10	9.8	4	21.2	1	27.5
High	10	9.3	1	22.0	1	21.5

The 11 February 2004 sampling (Table 3) highlights the transient nature of soil water repellency as affected by soil water content. The high treatment had become completely wettable while both the low and control treatments exhibited some degree of repellency, even though samples were collected on rainy days (11.9 mm on 10 February 2004 and 27.6 mm on 11 February 2004) following 8 days of fine weather (average temperature of 17 °C) and 1 month of below average rainfall. For both irrigation treatments, the soil had become completely wettable at the completion of the irrigation season (25 March 2004), while the soil of the Control treatment had retained extreme (severity class 10—Table 2) repellency.

These results suggest that by increasing the soil water content of these soils through effluent irrigation, the actual water repellency of the soil can be ameliorated. The changes in repellency observed between the start and the finish of the irrigation season are consistent with the widely accepted notion that water repellency is usually a transient feature of dry soils (e.g. Dekker and Ritsema, 2000; Doerr et al., 2000; Dekker et al., 2001; Keizer et al., 2005a). An irrigation strategy can be devised to eliminate water repellency at a desired time by very frequent irrigations, after which regular irrigations can help maintain soil wettability to optimise plant growth by retaining more water in the root zone. The observed water contents had a single-peaked distribution with the entire range of water contents, except the very wet range, well represented. Obviously, the control plots had the driest soil overall (Fig. 3). In contrast, the distribution of the severity of soil water repellency as a function of soil water content (% dry mass) was dichotomous for each treatment (Fig. 4). From Fig. 4 it appears that the irrigation water had not yet increased the

**Fig. 3 – Distribution of the observed soil water contents near the soil surface of the experimental site (Fig. 2) below the litter layer for three irrigation treatments (n = 54 for each treatment).****Fig. 4 – Severity of in situ soil water repellency (see Table 2) at the soil surface below the litter layer in relation to soil water content (% dry mass) in samples taken at the soil surface of the experimental site for three irrigation treatments (n = 54 for each treatment).**

degree of water-repellency as reported by Wallach et al. (2005): the wettest samples in severity class >8 are all in the control plots. The samples that are in the intermediate repellency severity classes are scattered over a wide range of water contents for each treatment, with the drier samples appearing in the control plots, which had the driest soil. For further analysis, we therefore pooled the samples from all treatments (n = 162). Most samples were classified as either extremely repellent (n = 72) or wettable (n = 58). It is interesting to note that the low repellency severity classes (class 2 and 3) are not represented at all.

Leighton-Boyce et al. (2005) also observed this dichotomous distribution of water repellency, and referred to another study with similar findings. They concluded that the transition from water repellent to wettable must be sudden. Dekker and Ritsema (1994) introduced the concept of a critical soil water content: a sharp threshold above which soils are wettable and below which soils are water repellent. However, in a later study, Dekker et al. (2001) revised this concept and replaced it with the concept of a “transition zone”. They proposed that, instead of occurring at a distinct soil water content threshold, the switch between water repellent and wettable conditions could occur at any point within a broader range of soil water contents. Within this transition zone, samples could be either repellent or wettable, with only wettable conditions existing above and repellent conditions existing below this zone. Dekker et al. (2001) based their conclusions on the actual water repellency and the water content of a large number of samples. According to Leighton-Boyce et al. (2005), Dekker et al.’s (2001) transition zone could be a consequence of

hysteretic soil water-repellency, with water repellency breaking down and re-establishing at different soil water contents. Alternatively, they suggest the transition zone might be an artefact of the scale discrepancy between water content and water repellency, as mentioned above. The latter explanation seems to be less likely, given the fact that our unconventional soil water content sampling technique did not affect the observed bimodality.

Experimental procedures involving drying of the samples before administering drops of water or ethanol solutions minimize hysteresis, yet variable degrees of water repellency are routinely reported. Even on the same soil sample, penetration times vary between droplets. We therefore argue that the severity of soil water repellency, hysteretic or non-hysteretic, varies considerably in space. In view of the spatial variation of every soil property (e.g. Jury et al., 1991, p. 269) this seems highly probable. The ensemble of spatially variable critical water contents with sharp thresholds then results in the transition zone reported by Dekker et al. (2001) for a population of soil samples. Strategies to manage water repellent soils should take this heterogeneity into consideration.

A sharp threshold of individual samples should result in few samples with intermediate values of water repellency, since those samples are exactly at their critical water content. Dekker et al. (2001) reported numerous samples in all of their seven WDPT classes of water repellency, possibly owing to their sample volume (50 cm³). Our data display this lack of intermediate ethanol values, with no samples at all in classes 2 and 3, and only 21 samples (13%) in classes 4 through 8 (Fig. 4). We therefore hypothesize that the hydrophobic substances produced by *E. globulus* create a sharp, possibly hysteretic, threshold for water repellency, but that spatial variation of soil properties and the distribution of these substances result in a wide bandwidth of critical water contents, in our plot between 10% and 37% dry mass.

An inverse relationship between repellency severity and soil water content has been reported by several authors (e.g. Dekker and Ritsema, 1994; Dekker et al., 1998; Doerr et al., 2000; Coelho et al., 2005; Keizer et al., 2005a; Leighton-Boyce et al., 2005). Our data (Fig. 4) are consistent with such relationship, with samples exhibiting extreme repellency predominately having low soil water contents, and wettable samples predominately having higher water contents.

Both the number of water repellent locations in the plantation and the severity of the repellency decrease with increasing water content, until the upper boundary of the “transition zone” is reached and the soil becomes completely wettable (Figs. 4 and 5). Using the inverse relationship between

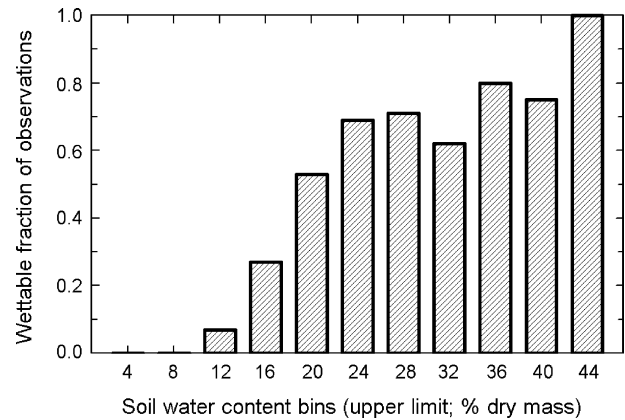


Fig. 5 – Fraction of samples taken at the soil surface below the litter layer of the experimental site that is wettable as a function of soil water content class (% dry mass).

water content and wettable fraction in Fig. 5, it is possible to estimate the fraction of the plantation soil surface that is wettable as a function of the spatially variable soil water content. If n water content observations are available, each representing a fraction $1/n$ of the total area of the field, the wettable fraction of each subarea can be read from Fig. 5. The wettable fraction of the field can be found by averaging. We did so for each treatment for each sampling round (18 samples per treatment), and found reasonable agreement in the dry and intermediate range (Table 4). The poorer performance in the wet range is presumably due to the small number of wet samples (11 samples wetter than 32% dry mass). Also, the area of the treatments was relatively small, and even in the random design spatial variation may have caused systematic differences between the samples from different rows. Still, the overall trends are captured well. For practical purposes of irrigation scheduling, the very dry range and the range at which the wettable fraction hardly increases with increasing wetness are of little interest. For this soil, the range between 4% and 36% dry mass water content is most relevant.

Water content is a parameter that should be monitored in any irrigation scheme, as it can be used to calculate and supply the correct amount of water to meet the water use as well as aid in preventing and/or combating the incidence of irregular wetting and preferential flow. Site specific information such as that in Fig. 5 will allow irrigation managers to estimate the fraction of the field that will be water repellent, based on just water content measurements. This facilitates determining the proper timing and amounts of irrigation to keep the soil water

Table 4 – Observed fraction of wettable samples and wettable portion of the soil surface estimated from Fig. 4 and water content observations for the various irrigation treatments (see Fig. 2 and main text) at three dates

Treatment	27 November 2003		11 February 2004		25 March 2004	
	Observed	Estimated	Observed	Estimated	Observed	Estimated
Control	0	0.18	0.33	0.42	0	0.05
Low	0	0.17	0.39	0.53	0.94	0.62
High	0	0.14	0.61	0.55	0.94	0.57

content above critical levels and thus avoid the onset of water repellent conditions.

Even though the approach presented here is dealing with the management of repellency at the immediate soil surface, it is important to note that water repellent conditions can exist at any depth in the soil profile, and that preferential flow could occur when these conditions are encountered. The success of the approach will depend on understanding site-specific conditions and adjusting irrigation accordingly. The application of this tool will require an initial systematic survey to determine the extent of water repellency throughout the soil profile of a plantation. If a water repellent layer is detected, both soil water content and repellency severity should be assessed from a representative area of the plantation. From this, the relationship between water content and the wettable fraction of the plantation can be calculated. Once determined, only soil water content, a parameter that must be monitored regularly in any irrigation scheme, is required.

It is interesting to note that a relatively high soil water content is required to ensure that 80% of the surface layer is wettable (32% dry mass; Fig. 5). This raises serious management considerations, particularly when dealing with the irrigation of effluent onto water repellent soils. In order to avoid excessive drainage and leaching of potentially harmful contaminants, it is important to maintain the soil water content of an effluent irrigated plantation low enough to prevent substantial drainage after rainfall. We used the Rosetta code (Schaap et al., 2001) to estimate from textural data and dry bulk densities the water content (% dry mass) in our soil at a pressure head of -333 cm to be approximately 0.04, rendering the entire field water-repellent. This would mean that the plantation is susceptible to excessive drainage and leaching via preferential flow, a common problem in water repellent soils (Dekker and Ritsema, 1994, 1995; Ritsema and Dekker, 1996). Further complications arise from the need to avoid salinization of the soil profile by a limited degree of over-irrigation (e.g. Bouwer, 1978, pp. 426–429). Navigating between keeping the soil wettable, reducing water loss through drainage, and preventing soil salinization clearly needs careful management. Further management techniques, such as the application of clay (e.g. Cann, 2000; Blackwell, 2000), may be necessary if the irrigation of effluent onto this plantation is to be sustainable.

4. Conclusions

This study assessed the in-field response of soil water content and soil water repellency to wetting by effluent irrigation in an established *E. globulus* plantation. We confirmed that there is a rapid transition between repellent and wettable conditions under established *E. globulus* plantation, but found that by increasing the soil water content through effluent irrigation, the water repellent nature of the soil was ameliorated. The soil from this study exhibits the commonly reported inverse relationship with samples exhibiting extreme repellency relating predominately to lower soil water contents and wettable samples relating predominately to higher water contents. At the field scale, the transition between repellent

and wettable conditions does not occur at a distinct point but rather it occurs somewhere between a soil water content of approximately 10% dry mass and 37% dry mass, even though the data suggest that the transition at the sampling scale occurs over a much narrower bandwidth. The relationship between soil water content and the percentage of the plantation area that is wettable can aid in preventing or combating the incidence of irregular wetting and preferential flow, by assisting the manager in predicting and supplying a sufficient amount of water to avoid the onset of water repellent conditions.

However, the interplay between reducing the leaching risk via preferential flow, and avoiding drainage and leaching during and after rainfall is complex, and needs careful management and/or new management techniques.

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REFERENCES

- Alyamani, M.S., Sen, Z., 1993. Determination of hydraulic conductivity from grain-size distribution curves. *Ground Water* 31, 551–555.
- Bisdorf, E.B.A., Dekker, L.W., Schoute, J.F.T., 1993. Water repellency of sieve fractions from sandy soils and relationships with organic material and soil structure. *Geoderma* 56, 105–118.
- Blackwell, P.S., 2000. Management of water repellency in Australia, and risks associated with preferential flow, pesticide concentration and leaching. *J. Hydrol.* 231–232, 384–395.
- Bond, R.D., Harris, J.R., 1964. The influence of the microflora on physical properties of soils. *Aust. J. Soil Res.* 2, 111–122.
- Bouwer, H., 1978. *Groundwater Hydrology*. McGraw-Hill Kogakusha Ltd., Tokyo.
- Bouwer, H., 2002. Artificial recharge of groundwater: hydrogeology and engineering. *Hydrogeol. J.* 10, 121–142.
- Briggs, D., 1977. *Sources and Methods in Geography: Soils*. Butterworths and Co (Publishers) Ltd, London, UK.
- Cann, M.A., 2000. Clay spreading on water repellent sands in the south east of South Australia—promoting sustainable agriculture. *J. Hydrol.* 231–232, 333–341.
- Coelho, C.O.A., Laouina, A., Regaya, K., Ferreira, A.J.D., Carvalho, T.M.M., Chaker, M., Naafa, R., Naciri, R., Boulet, A.K., Keizer, J.J., 2005. The impact of soil water repellency on soil hydrological and erosional processes under *Eucalyptus* and evergreen *Quercus* forest in the Western Mediterranean. *Aust. J. Soil Res.* 43, 309–318.
- DeBano, L.F., 1969. Water repellent soils: a worldwide concern in management of soil and vegetation. *Agric. Sci. Rev. Second Quart.* 11–18.

- DeBano, L.F., 2000. Water repellency in soils: a historical overview. *J. Hydrol.* 231–232, 4–32.
- Dekker, L.W., Ritsema, C.J., 1994. How water moves in a water repellent sandy soil. 1. Potential and actual water repellency. *Water Resour. Res.* 30, 2507–2518.
- Dekker, L.W., Ritsema, C.J., 1995. Fingerlike wetting patterns in two water-repellent loam soils. *J. Environ. Qual.* 24, 324–333.
- Dekker, L.W., Ritsema, C.J., 2000. Wetting patterns and moisture variability in water repellent Dutch soils. *J. Hydrol.* 231–232, 148–164.
- Dekker, L.W., Ritsema, C.J., Oostindie, K., Boersma, O.H., 1998. Effect of drying temperature on the severity of soil water repellency. *Soil Sci.* 163, 780–796.
- Dekker, L.W., Doerr, S.H., Oostindie, K., Ziogas, A.K., Ritsema, C.J., 2001. Water repellency and critical soil water content in a dune sand. *Soil Sci. Soc. Am. J.* 65, 1667–1674.
- Doerr, S.H., 1998. On standardizing the “Water drop penetration time” and the “Molarity of an ethanol droplet” techniques to classify soil hydrophobicity: a case study using medium textured soil. *Earth Surf. Processes Landforms* 23, 663–668.
- Doerr, S.H., Shakesby, R.A., Walsh, R.P.D., 1998. Spatial variability of soil hydrophobicity in fire-prone eucalyptus and pine forests, Portugal. *Soil Sci.* 163, 313–324.
- Doerr, S.H., Shakesby, R.A., Walsh, R.P.D., 2000. Soil water repellency: its causes, characteristics and hydro-geomorphological significance. *Earth-Sci. Rev.* 51, 33–65.
- Falkiner, R.A., Smith, C.J., 1997. Changes in soil chemistry in effluent-irrigated *Pinus radiata* and *Eucalyptus grandis* plantations. *Aust. J. Soil Res.* 35, 131–147.
- Feigin, A., Ravina, I., Shalhevet, J., 1991. Irrigation with treated sewage effluent. *Management for Environmental Protection. Advanced Series in Agricultural Sciences*, vol. 17. Springer-Verlag, Berlin.
- Franco, C.M.M., Clarke, P.J., Tate, M.E., Oades, J.M., 2000. Hydrophobic properties and chemical characterization of natural water repellent materials in Australian sands. *J. Hydrol.* 231–232, 47–58.
- Gee, G.W., Or, D., 2002. Particle-size analysis. In: Dane, J.H., Topp, G.C. (Eds.), *Methods of Soil Analysis. Part 4. Physical Methods*. Soil Science Society of America Inc., Madison, Wisconsin, USA, pp. 255–293.
- Harper, R.J., Gilkes, R.J., 1994. Soil attributes related to water repellency and the utility of soil survey for predicting its occurrence. *Aust. J. Soil Res.* 32, 1109–1124.
- Jury, W.A., Gardner, W.R., Gardner, W.H., 1991. *Soil Physics*, 5th ed. John Wiley & Sons Inc., New York.
- Keizer, J.J., Coelho, C.O.A., Matias, M.J.S., Domingues, C.S.P., Ferreira, A.J.D., 2005a. Soil water repellency under dry and wet antecedent weather conditions for selected land-cover types in the coastal zone of central Portugal. *Aust. J. Soil Res.* 43, 297–308.
- Keizer, J.J., Ferreira, A.J.D., Coelho, C.O.A., Doerr, S.H., Malvar, M.C., Domingues, C.S.P., Perez, I.M.B., Ruiz, C., Ferrari, K., 2005b. The role of tree stem proximity in the spatial variability of soil water repellency in a eucalypt plantation in coastal Portugal. *Aust. J. Soil Res.* 43, 251–259.
- Laidlaw, W.S., 2000. Blue Gum Effluent Irrigation Trial: Protocol for Irrigation Scheduling, Maintenance and Monitoring. Portland Aluminium, Portland, Victoria (unpubl.).
- Leighton-Boyce, G., Doerr, S.H., Shakesby, R.A., Walsh, R.P.D., Ferreira, A.J.D., Boulet, A.K., Coelho, C.O.A., 2005. Temporal dynamics of water repellency and soil moisture in eucalypt plantations, Portugal. *Aust. J. Soil Res.* 43, 269–280.
- Mallik, A.U., Rahman, A.A., 1985. Soil water repellency in regularly burned Calluna heathlands: comparison of three measuring techniques. *J. Environ. Manage.* 20, 207–218.
- McGhie, D.A., Posner, A.M., 1981. The effects of plant top material on the water repellence of fired sands and water repellent soils. *Aust. J. Agric. Res.* 32, 609–620.
- McKissock, I., Walker, E.L., Gilkes, R.J., Carter, D.J., 2000. The influence of clay type on reduction of water repellency by applied clays: a review of some West Australian work. *J. Hydrol.* 231–232, 323–332.
- Moore, G., Blackwell, P.S., 1998. Water repellence. In: Moore, G. (Ed.), *Soil Guide. Agriculture Western Australia*, pp. 53–63.
- Moral Garcia, F.J., Dekker, L.W., Oostindie, K., Ritsema, C.J., 2005. Water repellency under natural conditions in sandy soils of southern Spain. *Aust. J. Soil Res.* 43, 291–296.
- Morley, C.P., Mainwaring, K.A., Doerr, S.H., Douglas, P., Llewellyn, C.T., Dekker, L.W., 2005. Organic compounds at different depths in sandy soils and their role in water repellency. *Aust. J. Soil Res.* 43, 239–249.
- Morris, J.D., Wehner, B.A., 1987. Daily and Annual Water Use by Four Eucalypt Species Irrigated with Wastewater at Robinvale. Department of Conservation, Forests and Lands-Lands and Forestry Division, Victoria.
- Myers, B.J., Theiveyanathan, S., O'Brien, N.D., Bond, W.J., 1996. Growth and water use of *Eucalyptus grandis* and *Pinus radiata* plantations irrigated with effluent. *Tree Physiol.* 16, 211–219.
- Myers, B.J., Bond, W.J., Benyon, R.G., Falkiner, R.A., Polglase, P.J., Smith, C.J., Snow, V.O., Theiveyanathan, S., 1999. Sustainable Effluent-irrigated Plantations: An Australian Guideline. CSIRO Forestry and Forest Products, Canberra.
- Ritsema, C.J., Dekker, L.W., 1996. Water repellency and its role in forming preferred flow paths in soils. *Aust. J. Soil Res.* 34, 475–487.
- Ritsema, C.J., Dekker, L.W., 2000. Preface to a special issue: water repellency in soils. *J. Hydrol.* 231–232, 1–3.
- Ritsema, C.J., Dekker, L.W., 2005. Behaviour and management of water repellent soils—preface. *Aust. J. Soil Res.* 43, i–ii.
- Ritsema, C.J., van Dam, J.C., Dekker, L.W., Oostindie, K., 2005. A new modelling approach to simulate preferential flow and transport in water repellent porous media: model structure and validation. *Aust. J. Soil Res.* 43, 361–369.
- Roberts, F.J., 1966. The effects of sand type and fine particle amendments on the emergence and growth of of subterranean clover (*Trifolium subterraneum* L) with particular reference to water relations. *Aust. J. Agric. Res.* 17, 657–672.
- Schaap, M.G., Leij, F.J., van Genuchten, M.Th., 2001. ROSETTA: a computer program for estimating soil hydraulic parameters with hierarchical pedotransfer functions. *J. Hydrol.* 251, 163–176.
- Scott, D.F., 1992. The influence of vegetation type on soil wettability. In: Proceedings of the 17th Congress of the Soil Science Society of South Africa, University of Stellenbosch, Matieland, South Africa, 28 January–30 January, pp. 10B21–10B26.
- van Schilfgaarde, J., 1984. Drainage design for salinity control. In: Shainberg, I., Shalhevet, J. (Eds.), *Soil Salinity Under Irrigation. Processes and Management*. Springer-Verlag, Berlin, pp. 190–197.
- Wallach, R., Ben-Arie, O., Graber, E.R., 2005. Soil water repellency induced by long-term irrigation with treated sewage effluent. *J. Environ. Qual.* 34, 1910–1920.
- Xiong, X., Stagnitti, F., Turoczy, N.J., Allinson, G., Li, P., Nieber, L., Steenhuis, T.S., Parlange, J.Y., LeBlanc, M., Ziogas, A.K., Ferreira, A.J.D., Keizer, J.J., 2005a. Competitive sorption of metals in water repellent soils: implications for irrigation recycled water. *Aust. J. Soil Res.* 43, 351–356.
- Xiong, X., Stagnitti, F., Allinson, G., Turoczy, N.J., Li, P., LeBlanc, M., Cann, M.A., Doerr, S.H., Steenhuis, T.S., Parlange, J.Y., de Rooij, G., Ritsema, C.J., Dekker, L.W., 2005b. Effects of clay amendment on adsorption and desorption of copper in water repellent soils. *Aust. J. Soil Res.* 43, 397–402.